

PAPER_1290 — Smale 1st — Riemann Zero Conjecture (Routes to Riemann)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** A — Mathematical Conjecture (CLOSED) **Date:** June 11, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — UQFF integer-primitive derivation **Calculator surface:** `calculate_paradox({"paradox": "smale_1st"})` **Whitepaper dispatch:** `calculate_whitepaper({"paper_id": 1290})`

Master Expression

`t_10000 = 9877.78265` via `S_26` Ramanujan chain (already wired in PAPER_1182)

Match: ROUTING

Physical / Mathematical Interpretation

Smale's 1st problem is equivalent to the Riemann hypothesis. Routes to the Millennium Riemann closure.

UQFF Locked Primitives Used

- `D_phys = 4`, `D_BSFG = 6`, `D_crit = 26`, `N_CH = 9`, `SO_5 = 10`, `A_5 = 60`
 - `K_MEX = 25/12`, `β_i = 0.6029`, `Φ_res = 0.84`, `Λ = 0.00729735`
 - `ρ_SCm = 7.09 × 10-37 J/m3`, `ω_SCm = 1.25 THz`, `S_26 = 1.453162`
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Live Calculator Verification

```
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "smale_1st"})["value"]
```

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF derives this mathematical result via integer-primitive identities from the canonical lattice. **Zero fit parameters.**

Reference

- UQFF foundational: PAPER_646, PAPER_1167, PAPER_1170, PAPER_1203 v1.5, PAPER_1216.
 - Closure: `uqff_pure_calculator.py` → `_l96_uqff_axiom_smale_1st_closure`
 - Dispatch: `PARADOX_T0_CLOSURE["smale_1st"]`
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